

Reihaneh Vafadar

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Academic appointment

2025–present **Postdoctoral Fellow in Mathematics, McMaster University**
Advisors: Traian A. Pîrvu and Eric T. Sawyer

Education

2021–2025 **PhD in Mathematics, Université Laval**
Thesis: *Regularity and stochastic analysis for diffusion with divergence-free drifts*
Advisor: Damir Kinzebulatov

2010–2012 **MSc in Mathematics, Sharif University**
Thesis: *Dynamics and Pattern Formation in a Diffusive Predator–Prey System with Strong Allee Effect*
Advisor: M. Hesarakı

2006–2010 **BSc in Mathematics, Ferdowsi University**

Research interests

Partial differential equations; stochastic differential equations with singular or distributional drifts; regularity theory; stochastic control; mathematical finance.

Published articles

On divergence-free (form-bounded type) drifts.
Damir Kinzebulatov and Reihaneh Vafadar. *Discrete and Continuous Dynamical Systems – S*, 17 (5–6), 2083–2107. 2024.

Preprints

Optimal Investment with Switching Preferences.
Yu-Jui Huang, Liviu Ignat, Traian A. Pîrvu, and Reihaneh Vafadar. Preprint · arXiv:2607.22536. 2026.

SDEs with critical general distributional drifts: sharp solvability and blow-ups.
Damir Kinzebulatov and Reihaneh Vafadar. Preprint · arXiv:2506.09244. 2025.

Open problems in one-parameter operator semigroups theory.
Kodjo Raphaël Madou, Ivan Remizov, and Reihaneh Vafadar. Preprint · arXiv:2410.00416. 2024.

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Academic CV · October 2026

Working manuscripts

On the Quantitative Properties of Optimal Investment with Switching Preferences.

Traian A. Pirvu and Reihaneh Vafadar. Working manuscript. 2026.

Asymptotic Analysis of Optimal Investment with Switching Preferences.

Traian A. Pirvu and Reihaneh Vafadar. Working manuscript. 2026.

Teaching experience

2025–2026	Instructor, MATH 3IA3: Introduction to Real Analysis McMaster University Fall 2025: Instructor, Section C02. Winter 2026: Instructor and sole course coordinator for the only section; coordination of two teaching assistants. Fall 2026: Instructor, Section C02.
2023–2025	Mathematics tutor, Université Laval Mathematics Help Centre (CDA), drop-in tutoring.
2021–2025	Grader, Université Laval Examinations and homework assignments.
2011–2012	Teaching assistant, Sharif University Calculus I and II.

Selected talks

January 2026	Postdoc Threads, McMaster University, Hamilton.
June 2025	Canadian Mathematical Society Summer Meeting, Québec City.
December 2023	Canadian Mathematical Society Winter Meeting, Montréal.

Selected awards and support

2021, 2023	Prize for the Valorization of Student Publications, CIMMUL, Université Laval.
2021–2022	Research scholarship, Institut des Sciences Mathématiques (ISM), Québec.

Skills and languages

Technical skills: Python, LaTeX.

Languages: English, French, Persian.
